

98

ξ	-1	0	1	2
p	0,3	0,4	0,2	0,1

$$E\xi = \sum_{i=1}^4 x_i \cdot p_i = 0,3(-1) + 0(0,4) + 1 \cdot 0,2 + 2 \cdot 0,1 = 0,1$$

$$D\xi = E\xi^2 - (E\xi)^2 = \sum_{i=1}^4 x_i^2 \cdot p_i - (E\xi)^2 = 1 \cdot 0,3 + 1 \cdot 0,2 + 4 \cdot 0,1 - 0,1^2 = 0,89$$

$$\sigma_\xi = \sqrt{D\xi} = \sqrt{0,89}$$

Омбер: $E\xi = 0,1$ $D\xi = 0,89$ $\sigma_\xi = \sqrt{0,89}$

99

ξ	-2	-1	0	1	3
p	0,4	0,2	0,1	p_n	0,2

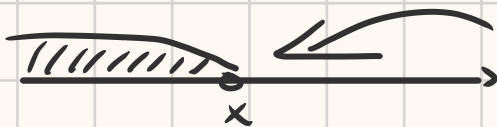
$$\sum p_i = 1 \quad p_n = 1 - 0,4 - 0,2 - 0,1 - 0,2 = 0,1$$

$$E\xi = -0,8 - 0,2 + 0,1 + 0,6 = -0,3$$

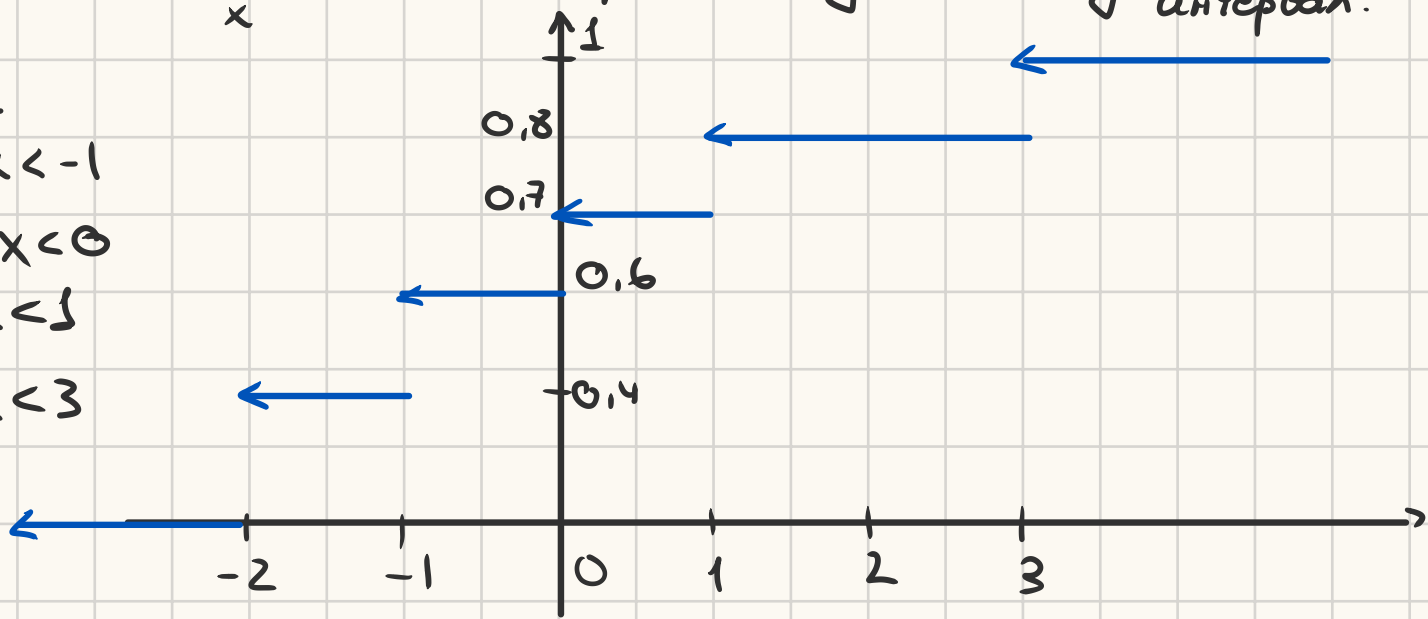
$$D\xi = 4 \cdot 0,4 + 1 \cdot 0,2 + 1 \cdot 0,1 + 9 \cdot 0,2 - 0,09 = 3,61$$

$$\sigma_\xi = \sqrt{D\xi} = 1,9$$

$$P(0 \leq \xi < 2) = p(\xi=0) + p(\xi=1) = 0,2$$

$F(x) = P(\xi < x)$  Вер-ть попадания в данный интервал.

$$F(x) = \begin{cases} 0, & x < -2 \\ 0,4, & -2 \leq x < -1 \\ 0,6, & -1 \leq x < 0 \\ 0,7, & 0 \leq x < 1 \\ 0,8, & 1 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$



Омбер: $E\xi = -0,3$

$$D\xi = 3,61$$

$$\sigma_\xi = 1,9$$

$$p(0 \leq \xi < 2) = 0,2$$

$$F(x) = \begin{cases} 0, & x < -2 \\ 0,4, & -2 \leq x < -1 \\ 0,6, & -1 \leq x < 0 \\ 0,7, & 0 \leq x < 1 \\ 0,8, & 1 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

<u>1001</u>	ξ	1	4	5	6	8
	p	0.2	0.1	p_n	0.3	0.3

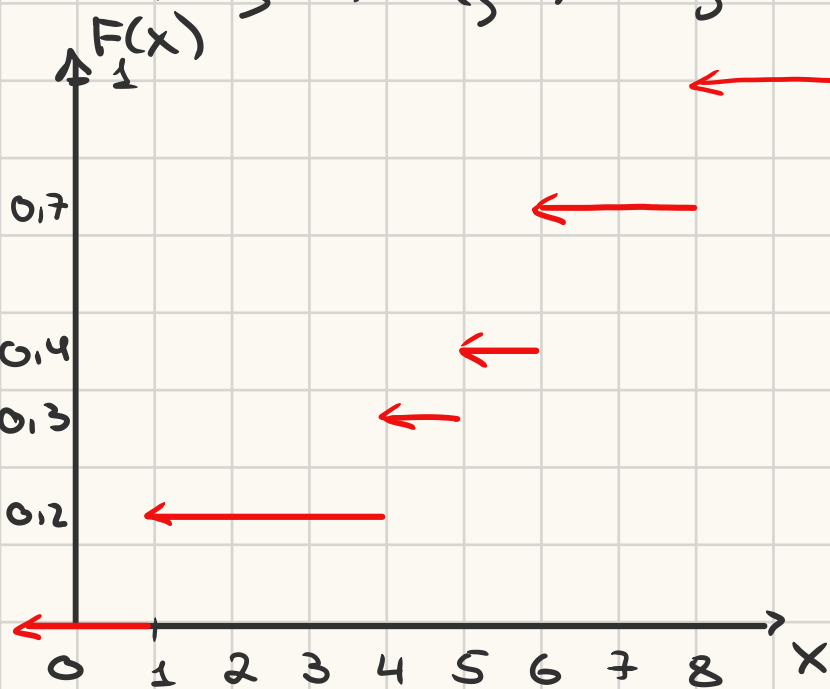
$$\sum p_i = 1 \quad p_n = 1 - 0.9 = 0.1$$

$$E\xi = 0.2 + 0.4 + 0.5 + 1.8 + 2.4 = 5.3$$

$$D\xi = 0.2 + 1.6 + 2.5 + 10.8 + 19.2 - (5.3)^2 = 6.21$$

$$\sigma_\xi = \sqrt{6.21} \approx 2.4920$$

$$P(1 < \xi \leq 6) = P(\xi=4) + P(\xi=5) + P(\xi=6) = 0.5$$



$$F(x) = \begin{cases} 0, & x \leq 1 \\ 0.2, & 1 < x \leq 4 \\ 0.3, & 4 < x \leq 5 \\ 0.4, & 5 < x \leq 6 \\ 0.7, & 6 < x \leq 8 \\ 1, & x > 8 \end{cases}$$

1021 ξ и η - независимы $E\xi = -1$ $D\xi = 2$

$$E\eta = 2 \quad D\eta = 5$$

$$E(3\xi - 2\eta) = E(3\xi) - E(2\eta) = 3(-1) + (-2) \cdot 3 = -9$$

$$D(3\xi - 2\eta) = D(3\xi) - D(2\eta) = 9 \cdot 2 + 4 \cdot 5 = 38$$

Ответ: $E(3\xi - 2\eta) = -9$
 $D(3\xi - 2\eta) = 38$

1031 $E\xi = 2$ $E\eta = 1$ $\text{cov}(\xi, \eta) = -1$
 $D\xi = 1$ $D\eta = 3$

$$E(4\xi - \eta) = E(4\xi) - E(\eta) = 4E(\xi) - E(\eta) = 7$$

$$D(4\xi - \eta) = D(4\xi) - D(\eta) + 2\text{cov}(4\xi, \eta) = 16 + 3 + 8 = 27$$

<u>1071</u>	ξ	0	1	2
	p	$1/10$	$6/10$	$3/10$

$$0 \leq \xi \leq 2$$

$$E\xi = 0.6 + 0.6 = 1.2$$

$$D\xi = 0.6 + 1.2 - (1.2)^2 = 0.36$$

$$\sigma_\xi = \sqrt{D\xi} = \sqrt{1.8} = 1.34$$

$$p(\xi=0) = \frac{C_2^0 \cdot C_3^3}{C_5^3} = \frac{1}{10}$$

$$p(\xi=2) = \frac{C_2^2 \cdot C_3^1}{C_5^3} = \frac{3}{10}$$

$$p(\xi=1) = \frac{C_2^1 \cdot C_3^2}{C_5^3} = \frac{6}{10}$$

1081

ξ	0	1	2	3
p	---	- + - + - - - - +	+ + - + - + - + +	+ + +

$$p(A_1) = 0.8$$

$$p(A_2) = 0.6$$

$$p(A_3) = 0.5$$

Варианты событий

вер.-и. $p(\xi=0) = 0.04$ $p(\xi=1) = 0.26$ $p(\xi=2) = 0.46$ $p(\xi=3) = 0.24$

$$E\xi = 0.26 + 2 \cdot 0.46 + 3 \cdot 0.24 = 1.9$$

$$D\xi = 0.26 + 4 \cdot 0.46 + 9 \cdot 0.24 - (1.9)^2 = 0.65$$

1091

ξ	1	2	3	4
p	0.5	0.25	0.125	0.125

$$p(\xi=1) = 0.5$$

$$p(\xi=2) = 0.5 \cdot 0.5 = 0.25$$

$$p(\xi=3) = (0.5)^3 = 0.125$$

$$p(\xi=4) = q^3 \cdot p + q^4 = 0.125$$

$$E\xi = 0.5 + 0.5 + 0.375 + 0.5 = 1.875$$

$$D\xi = 0.5 + 4 \cdot 0.25 + 9 \cdot 0.125 + 16 \cdot 0.125 - (1.875)^2 = 1.109375$$

1101

ξ	0	1	2	3	4
p	$(0.4)^4$	$4(0.6)(0.4)^3$	$6(0.6)^2(0.4)^2$	$4(0.6)^3(0.4)$	$(0.6)^4$

$$E\xi = 4(0.6)(0.4)^3 + 6(0.6)^2(0.4) \cdot 2 + 3 \cdot 4(0.6)^3(0.4) + 4 \cdot (0.6)^4 = 2.4$$

$$D\xi = E\xi^2 - (2.4)^2 = 0.96$$

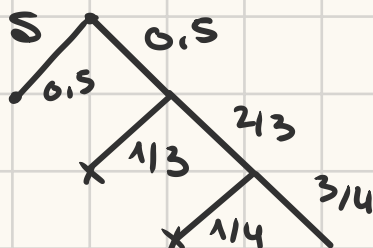
1111

ξ	2-8	9	10	11	12
p	$\frac{26}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

$$E\xi = 0 \cdot \frac{26}{36} + 50 \cdot \frac{4}{36} + 80 \cdot \frac{3}{36} + 90 \cdot \frac{2}{36} + 100 \cdot \frac{1}{36} = 20 \text{ pyдней}$$

Омбер: 20 pyдней

1121



ξ	1	-1	1	-1	...	$(-1)^{n-1}$
p	$\frac{1}{2}$	$\frac{1}{2} \cdot \frac{1}{3}$	$\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4}$	$\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{5}$	...	$\frac{1}{n(n+1)}$

$$E\xi = \sum_{k=1}^{\infty} (-1)^{k-1} \cdot \frac{1}{n(n+1)} = \sum_{k=1}^{\infty} (-1)^{k-1} \cdot \frac{1}{n} - (-1)^{k-1} \cdot \frac{1}{n+1} \quad \text{①}$$

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1} \xrightarrow{n \rightarrow \infty} 1$$

$$\Leftrightarrow \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} + \sum_{n=1}^{\infty} (-1)^{(n+1)-1} \frac{1}{n+1} = 2 \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n} - 1 \Leftrightarrow$$

$$\ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3} - \dots (-1)^n \frac{x^n}{n} \Leftrightarrow 2 \ln(2) - 1 \approx 0.386$$

$$D\xi = E\xi^2 - (E\xi)^2 = 1 - (0.386)^2 = 0.851$$

Ответ: $E\xi = 0.386$

$D\xi = 0.851$

1151

$$I_i = \begin{cases} 1, & \text{если } i\text{-ю} \\ 0, & \text{иначе} \end{cases}$$

$$P(I_i = 1) = \underbrace{\frac{1}{2}}_{\text{сеча}} \cdot \underbrace{\frac{1}{2}}_{\text{сраба}} = \frac{1}{4} \quad E I_i = \frac{1}{4}$$

$$\xi = I_1 + \dots + I_{100} \Rightarrow E\xi = \sum_{i=1}^{100} E I_i = 100 \cdot \frac{1}{4} = 25$$

Ответ: 25